

Inappropriate Shocks From Subcutaneous vs Transvenous Implantable Cardioverter-Defibrillators



Individual Participant Data Meta-Analysis of Randomized Trials

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ABSTRACT

BACKGROUND The PRAETORIAN (Prospective Randomized Comparison of Subcutaneous and Transvenous Implantable Cardioverter Defibrillator Therapy) and ATLAS (Avoid Transvenous Leads in Appropriate Subjects) trials demonstrated a reduction in lead-related complications with subcutaneous compared with transvenous implantable cardioverter-defibrillators (ICDs) in patients with standard ICD indications. However, they were not individually powered to assess differences in inappropriate shock.

OBJECTIVES The aims of this study were to compare the rates of a first inappropriate shock between subcutaneous and transvenous ICDs and to explore the underlying causes.

METHODS An individual participant data meta-analysis of PRAETORIAN and ATLAS was performed, using Cox regression models stratified by trial. The primary endpoint was time to first inappropriate shock.

RESULTS The pooled modified intention-to-treat population included 1,342 patients (median age 58 years, 21.8% women, 77.7% with primary prevention indications) who had been randomized to subcutaneous or transvenous ICDs. A proprietary algorithm aimed at reducing inappropriate shock was activated at baseline or at some point during follow-up in 77.1% of patients with subcutaneous ICDs. During a median follow-up period of 3.5 years, a first inappropriate shock occurred more frequently in patients randomized to subcutaneous compared with transvenous ICDs (2.5 vs 1.5 per 100 patient-years; HR: 1.61; 95% CI: 1.06-2.45; $P = 0.03$). Patients with subcutaneous ICDs had a higher hazard of first inappropriate shock due to cardiac oversensing (HR: 15.07; 95% CI: 3.60-63.15; $P < 0.001$) and electromagnetic interference or myopotentials (HR: 8.19; 95% CI: 1.88-35.64; $P = 0.005$) but a lower hazard of first inappropriate shock due to atrial arrhythmia (HR: 0.37; 95% CI: 0.19-0.71; $P = 0.003$).

CONCLUSIONS Although first inappropriate shocks were infrequent overall, they were more common with subcutaneous ICDs. First inappropriate shocks due to cardiac oversensing and electromagnetic interference were more common with subcutaneous ICDs, while first shocks due to atrial arrhythmia were more common with transvenous ICDs. (JACC. 2026;87:1732-1743) © 2026 by the American College of Cardiology Foundation.

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The implantable cardioverter-defibrillator (ICD) is a guideline-recommended therapy in patients with indications for primary or secondary prevention of sudden cardiac death.^{1,2} Transvenous ICD systems consist of a pulse generator and a lead reaching the heart via the cephalic, subclavian, or axillary vein. Endovascular lead placement carries a risk for perioperative and chronic lead-related complications.³

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The subcutaneous ICD was developed to overcome some of the limitations related to endovascular ICD lead placement.⁴ Two randomized trials demonstrated a reduction in lead-related complications with subcutaneous compared with transvenous ICDs in patients with primary or secondary prevention indications for ICDs without indications for pacing.⁵⁻⁷ However, neither trial was powered to assess ICD shock efficacy or the risk for inappropriate shock. Historically, rates of inappropriate shock in patients undergoing ICD implantation exceeded 5 per 100 patient-years.⁸ Reducing inappropriate shocks in ICD recipients is of considerable interest because they are painful and associated with impaired quality of life, proarrhythmia, and accelerated battery depletion. Earlier trials in patients undergoing transvenous ICD implantation found an association of inappropriate shock with increased mortality.^{9,10}

In this pooled analysis of the PRAETORIAN (Prospective Randomized Comparison of Subcutaneous and Transvenous Implantable Cardioverter Defibrillator Therapy) and ATLAS (Avoid Transvenous Leads in Appropriate Subjects) trials, we aimed to assess the effect of subcutaneous compared with transvenous ICDs on receiving a first inappropriate shock and to explore the underlying causes. Although both PRAETORIAN and ATLAS advanced our understanding of the safety and effectiveness of subcutaneous and transvenous ICD systems, each trial alone was underpowered to fully assess differences in inappropriate shock rates. This individual participant data meta-analysis combines the 2 largest randomized trials to date, providing the most comprehensive and statistically robust evaluation of inappropriate shock burden across modern ICD platforms. By analyzing patient-level data, stratifying mechanisms of inappropriate shock, and examining device-specific and patient-level contributors, this study offers a definitive reference point for clinicians and health systems weighing the trade-offs of extravascular vs transvenous defibrillation.

METHODS

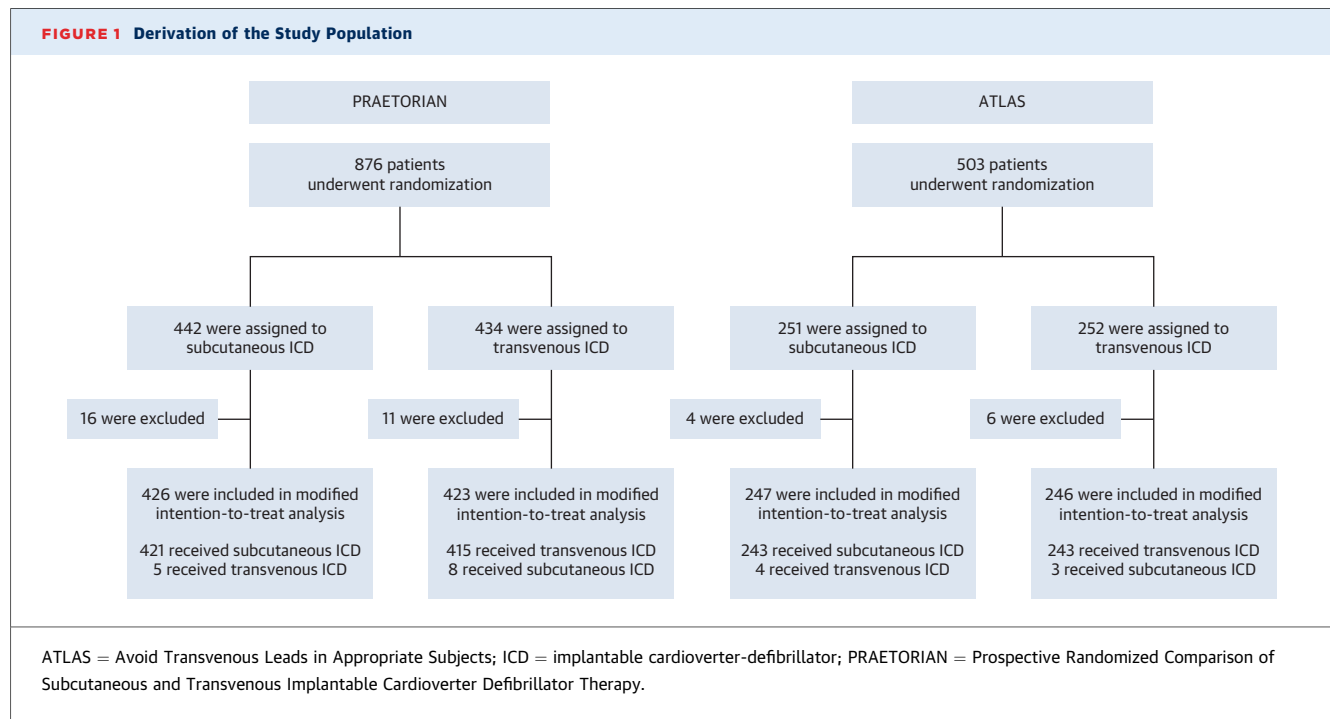
STUDY POPULATION. We created a central database harmonizing individual participant-level data of the PRAETORIAN and ATLAS trials. The study protocols and the main results of the 2 trials are published elsewhere.^{5,7,11,12} Planning to pool the data began in 2021 (after publication of the PRAETORIAN results and before publication of the ATLAS results). The main eligibility criteria were similar, as both trials enrolled patients with primary or secondary indications for ICDs without indications for pacing. Both trials randomized study participants to receive either subcutaneous or transvenous ICDs. Programming of all devices was done according to the most up-to-date evidence available at the time of initiation of the individual trials.^{11,12} In PRAETORIAN, the cutoff for the fast ventricular tachycardia zone in patients with primary prevention indications was set at 180 beats/min in patients with subcutaneous ICDs and 182 beats/min (or as close to 182 beats/min as possible, depending on the device manufacturer) in patients with transvenous ICDs.¹³ In ATLAS, this cutoff was set at 200 beats/min in both groups.¹⁴ The cutoff for the ventricular fibrillation zone was set at 250 beats/min in both subcutaneous and transvenous ICDs in both trials. Antitachycardia pacing was turned on in transvenous ICDs in the fast ventricular zone in both PRAETORIAN and ATLAS. Additional details on device programming are published elsewhere.^{5,7,11,12} Approvals from local research ethics boards were obtained prior to initiation of the trials, and all study participants provided informed consent.

PRAETORIAN enrolled participants at 39 centers in Europe and the United States between 2011 and 2017. The primary outcome was a composite of device-related complications or inappropriate shock. The trial met its primary endpoint, demonstrating noninferiority of the subcutaneous ICD compared with the transvenous ICD (HR: 0.99; 95% CI: 0.71-1.39; $P = 0.01$ for noninferiority). There was a numerical increase in inappropriate shock with subcutaneous vs transvenous ICDs (HR: 1.43; 95% CI: 0.89-2.30) that did not meet statistical significance.⁵

ATLAS enrolled participants at 14 centers in Canada between 2017 and 2021. The primary outcome was perioperative major lead-related complications. The trial met its primary endpoint, showing a reduction in perioperative major lead-related complications with the subcutaneous ICD compared with the transvenous ICD (0.4% vs 4.8%; $P = 0.001$). Similar to the PRAETORIAN trial, there also was a

ABBREVIATIONS AND ACRONYMS

ICD = implantable
cardioverter-defibrillator



numerical but statistically nonsignificant increase in inappropriate shock with subcutaneous ICDs vs transvenous ICDs (HR: 2.37; 95% CI: 0.98-5.77).⁷

During the conduct of the 2 trials, a proprietary algorithm that was developed to reduce inappropriate shock due to cardiac oversensing with the subcutaneous ICD (SMART Pass, Boston Scientific) was introduced (Supplemental Figure 1). This feature was available through a firmware upgrade for the second-generation model (A209 EMBLEM) and was built in to the third-generation model (A219 EMBLEM MRI). The algorithm uses a high-pass filter at 8 to 9 Hz to attenuate low-frequency signals (eg, T wave), thereby enhancing the QRS/T wave ratio and improving sensing accuracy.¹⁵

ENDPOINTS. Both PRAETORIAN and ATLAS adjudicated ICD therapies centrally.^{5,7} Adjudicators were not blinded to randomized treatment. The primary endpoint of this meta-analysis was time to first inappropriate shock as adjudicated in the individual trials, defined as shock in the absence of ventricular tachycardia or ventricular fibrillation. A shock for ventricular tachycardia below the therapy zone with oversensing of cardiac signals was not classified as inappropriate.⁵ The secondary endpoint was time to first inappropriate ICD therapy, defined as anti-tachycardia pacing or shock in the absence of ventricular tachycardia or ventricular fibrillation. Reasons for inappropriate shock or inappropriate ICD

therapy were classified into 3 adjudicated categories: 1) cardiac oversensing, defined as P-wave or T-wave oversensing; 2) electromagnetic interference or myopotentials; and 3) atrial arrhythmia, defined as atrial fibrillation or supraventricular tachycardia (including sinus tachycardia). Sample tracings are shown in Supplemental Figure 2.

STATISTICAL ANALYSIS. The statistical analysis plan for this report was finalized in May 2025. The choice of the primary endpoint, the primary analysis model, and all subgroup analyses were specified prior to initiation of any analyses. The analyses were performed between May and August 2025. All analyses were performed primarily in a modified intention-to-treat population (ie, patients were analyzed in the groups they were randomly assigned to; patients who did not receive any device, were lost to follow-up, withdrew consent before the first successful device implantation, or underwent randomization in error were excluded). Baseline characteristics of study participants are presented for the overall pooled cohort, according to assigned device group and trial, and separately for participants with and without a first inappropriate shock. Continuous variables are expressed as median (Q1-Q3), and categorical variables are expressed as count (percentage). The date of the first successful device implantation following randomization (irrespective of assigned group) was considered the reference date for all outcome

TABLE 1 Pooled Baseline Characteristics of Participants Enrolled in the PRAETORIAN and ATLAS Trials

	Total (N = 1,342)	Subcutaneous ICD (n = 673)	Transvenous ICD (n = 669)
Follow-up duration, y	3.5 (2.5-4.5)	3.5 (2.5-4.4)	3.6 (2.5-4.6)
Age, y	58.2 (50.0-67.0)	57.8 (49.0-66.0)	59.0 (51.0-67.0)
Female	293 (21.8)	148 (22.0)	145 (21.7)
Body mass index, ^a kg/m ²	27.5 (24.7-31.4)	27.2 (24.5-31.1)	27.8 (25.0-31.7)
ICD indication			
Primary prevention	1,043 (77.7)	521 (77.4)	522 (78.0)
Secondary prevention	299 (22.3)	152 (22.6)	147 (22.0)
Diagnosis ^b			
Ischemic cardiomyopathy	766 (57.1)	375 (55.7)	391 (58.5)
Nonischemic cardiomyopathy	310 (23.1)	153 (22.7)	157 (23.5)
Other	284 (21.2)	152 (22.6)	132 (19.7)
Genetic arrhythmic syndrome	80 (6.0)	41 (6.1)	39 (5.8)
Hypertrophic cardiomyopathy	113 (8.4)	59 (8.8)	54 (8.1)
Idiopathic ventricular fibrillation	99 (7.4)	58 (8.6)	41 (6.1)
Congenital heart disease	7 (0.5)	3 (0.5)	4 (0.6)
Other diagnosis	8 (0.6)	6 (0.9)	2 (0.3)
Left ventricular ejection fraction ^c			
≤35%	769 (68.7)	380 (68.3)	389 (69.1)
>35%	350 (31.3)	176 (31.7)	174 (30.9)
Hypertension	639 (47.6)	312 (46.4)	327 (48.9)
Diabetes mellitus	335 (25.0)	161 (23.9)	174 (26.0)
Medication at baseline			
Beta-blockers	1,096 (81.7)	550 (81.7)	546 (81.6)
Antiarrhythmic drugs	92 (6.9)	52 (7.7)	40 (6.0)
Region (study center)			
North America	553 (41.2)	279 (41.5)	274 (41.0)
Europe	789 (58.8)	394 (58.5)	395 (59.0)
Model implanted in patients randomized to subcutaneous ICDs ^d			
SQ-RX model 1010 (first generation)		183 (27.2)	
A209 EMBLEM (second generation)		208 (30.9)	
A219 EMBLEM MRI (third generation)		273 (40.6)	
Crossed over to transvenous ICD		9 (1.4)	

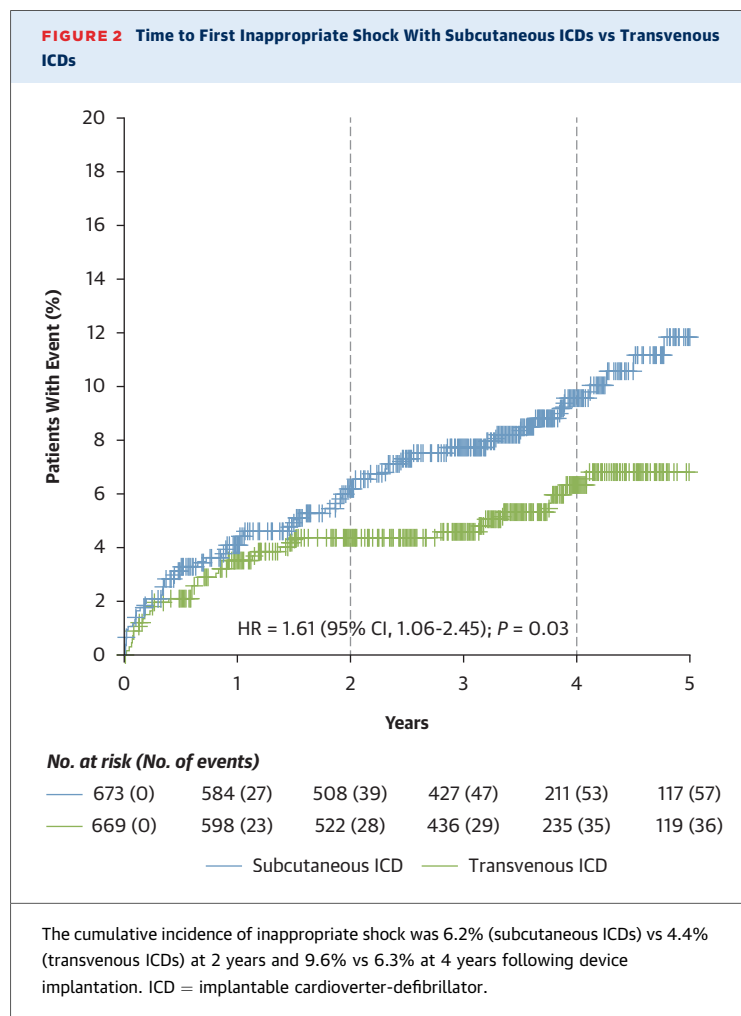
Values are median (Q1-Q3) or n (%). ^aBody mass index was missing for 4 patients (0.3%). ^bDiagnosis categories are not mutually exclusive. ^cLeft ventricular ejection fraction was missing for 223 patients (16.6%). ^dNine patients randomized to receive subcutaneous ICDs crossed over to transvenous ICDs, and 11 patients randomized to receive transvenous ICDs crossed over to subcutaneous ICDs. The first-generation model was manufactured by Cameron Health/Boston Scientific, and the second- and third-generation models were manufactured by Boston Scientific.

ATLAS = Avoid Transvenous Leads in Appropriate Subjects; ICD = implantable cardioverter defibrillator; PRAETORIAN = Prospective Randomized Comparison of Subcutaneous and Transvenous Implantable Cardioverter Defibrillator Therapy.

analyses. The count and percentage of participants who experienced the primary or secondary endpoint are presented. Event rates are reported per 100 patient-years. Cox regression models stratified by trial were used to assess the relative hazard of the primary and secondary endpoints in patients randomized to receive a subcutaneous vs transvenous ICD, respectively, with HRs and 95% CIs reported. The assumption of proportional hazards was checked by adding an interaction term with follow-up time. The burden of inappropriate shock during follow-up in participants who had at least 1 inappropriate shock is expressed as median (Q1-Q3) and visualized using bar plots. Exploratory analyses aimed at

identifying predictors of a first inappropriate shock were done using univariable and multivariable (variable included if $P \leq 0.20$ on univariable analysis) Cox regression models. Subgroup analyses according to age, sex, left ventricular ejection fraction, and baseline use of a beta-blocker and/or an antiarrhythmic drug were conducted using Cox regression models with an interaction term. Kaplan-Meier curves were used for visualization.

SMART PASS ANALYSES. In patients randomized to receive subcutaneous ICDs and who received first inappropriate shocks, the proportion of patients with events before or in the absence of a first activation of the SMART Pass algorithm is reported. Additional



Cox regression models including first activation of the algorithm (ie, at baseline or during follow-up) as a time-dependent covariate were built. Furthermore, the rates of first inappropriate shock without or before and with or after SMART Pass activation are reported.

SENSITIVITY ANALYSES. The following post hoc sensitivity analyses were conducted. First, for the primary endpoint of time to first inappropriate shock, results are presented separately for the 2 trials. In addition, formal testing for interaction with trial was done. Second, the analyses for the primary and secondary endpoint were repeated in the per-protocol and as-treated populations. Third, restricted mean survival time analyses until first inappropriate shock were performed, with the between-group difference reported. Fourth, analyses accounting for the competing risk for death using the Fine-Gray method

were done, with subdistribution hazards and 95% CIs reported. Fifth, for the primary endpoint of time to first inappropriate shock, analyses stratified by subcutaneous ICD device generation received at baseline were performed.

A 2-sided P value <0.05 was considered to indicate statistical significance. All analyses were performed using SAS version 9.4.

RESULTS

The pooled modified intention-to-treat population included 1,342 patients (Figure 1). Of these, 673 (50.1%) and 669 (49.9%) had been randomized to receive subcutaneous and transvenous ICDs, respectively. Of those randomized to subcutaneous ICDs, 183 (27.2%) received first-generation devices, 208 (30.9%) received second-generation devices, and 273 (40.6%) received third-generation devices. Before or during the implantation procedure, 9 patients randomized to subcutaneous ICDs crossed over to transvenous ICDs, and 11 patients randomized to transvenous ICDs crossed over to subcutaneous ICDs.

Overall, the median age of patients was 58.2 years, and 293 (21.8%) were women (Table 1). The majority of patients (77.7%) had primary prevention indications. A total of 766 (57.1%) had ischemic cardiomyopathy, 310 (23.1%) had nonischemic cardiomyopathy, and 284 (21.2%) had other underlying diagnoses, including a genetic arrhythmic syndrome, hypertrophic cardiomyopathy, or idiopathic ventricular fibrillation. The proportion of patients with left ventricular ejection fractions $\leq 35\%$ was 68.7%. In total, 1,096 (81.7%) and 92 (6.9%) patients reported baseline use of beta-blockers and antiarrhythmic drugs, respectively. Baseline characteristics stratified by trial are presented in Supplemental Table 1.

INAPPROPRIATE SHOCK. During a median follow-up period of 3.5 years (Q1-Q3: 2.5-4.5 years), first inappropriate shocks occurred in 57 patients (8.5%) randomized to receive subcutaneous ICDs and in 36 (5.4%) randomized to transvenous ICDs. Corresponding event rates were 2.5 and 1.5 per 100 patient-years, respectively (HR: 1.61; 95% CI: 1.06-2.45; $P = 0.03$) (Figure 2). The cumulative incidence of inappropriate shock was 6.2% vs 4.4% at 2 years and 9.6% vs 6.3% at 4 years following device implantation. Results stratified by trial are presented in Supplemental Table 2. There was no interaction with trial (P for interaction = 0.34). Patients with subcutaneous ICDs had a higher hazard of a first

TABLE 2 Time to First Inappropriate Shock With Subcutaneous vs Transvenous ICDs

	Subcutaneous ICD (n = 673)		Transvenous ICD (n = 669)			
		Event Rate		Event Rate	HR (95% CI)	P Value
Inappropriate shock	57 (8.5)	2.5	36 (5.4)	1.5	1.61 (1.06-2.45)	0.03
Reason for inappropriate shock						
Cardiac oversensing	29 (4.3)	1.2	2 (0.3)	0.1	15.07 (3.60-63.15)	<0.001
Electromagnetic interference or myopotentials	16 (2.4)	0.7	2 (0.3)	0.1	8.19 (1.88-35.64)	0.005
Atrial arrhythmia	12 (1.8)	0.5	32 (4.8)	1.4	0.37 (0.19-0.71)	0.003

Values are n (%) unless otherwise indicated. Event rates are reported per 100 patient-years. HRs are based on the comparison of subcutaneous vs transvenous ICDs. The assumption of proportional hazards was not violated (*P* = 0.74). In the PRAETORIAN trial, there were 11 patients randomized to subcutaneous ICDs who had shocks for ventricular tachycardia below the therapy zone with oversensing of cardiac signals. These shocks were not classified as inappropriate. There were no such events in the ATLAS trial.

Abbreviations as in [Table 1](#).

inappropriate shock due to cardiac oversensing (HR: 15.07; 95% CI: 3.60-63.15) and electromagnetic interference or myopotentials (HR: 8.19; 95% CI: 1.88-35.64) than patients with transvenous ICDs (Table 2, Figure 3). Patients with subcutaneous ICDs had a lower hazard of a first inappropriate shock due to atrial arrhythmia than patients with transvenous ICDs (HR: 0.37; 95% CI: 0.19-0.71).

Baseline characteristics of patients with and without inappropriate shock were largely similar (Supplemental Table 3). In adjusted analyses, only randomization to a subcutaneous ICD was associated with the hazard of first inappropriate shock due to cardiac oversensing, electromagnetic interference, or myopotentials (adjusted HR: 11.86; 95% CI: 4.27-32.98) (Table 3). Randomization to a subcutaneous ICD was associated with a lower hazard (adjusted HR: 0.40; 95% CI: 0.20-0.80), and nonischemic cardiomyopathy (adjusted HR: 3.03; 95% CI: 1.54-5.95) and enrollment in the PRAETORIAN trial (adjusted HR: 3.17; 95% CI: 1.04-9.70) were associated with a higher hazard of first inappropriate shock due to atrial arrhythmia. We did not identify predictors of a first inappropriate shock for any reason within the strata of patients randomized to subcutaneous and transvenous ICDs, respectively (Supplemental Table 4).

In patients with first inappropriate shock, the median number of inappropriate shocks during follow-up was 1 (Q1-Q3: 1-2) for those randomized to subcutaneous ICDs and 1 (Q1-Q3: 1-3) for those randomized to transvenous ICDs. A total of 36 patients (63.2%) with subcutaneous ICDs and 19 patients (52.8%) with transvenous ICDs who had first inappropriate shocks did not have recurrent inappropriate shock during follow-up (Figure 4). Eight patients (14.0%) with subcutaneous ICDs and 6 patients (16.7%) with transvenous ICDs had 2 first

inappropriate shocks, and 4 (7.0%) vs 5 (13.9%) and 3 (5.3%) vs 1 (2.8%) had 3 and 4 inappropriate shocks, respectively. Finally, there were 6 patients (10.5%) with subcutaneous ICDs and 5 patients (13.9%) with

FIGURE 3 Reasons for a First Inappropriate Shock in Patients With Subcutaneous vs Transvenous ICDs

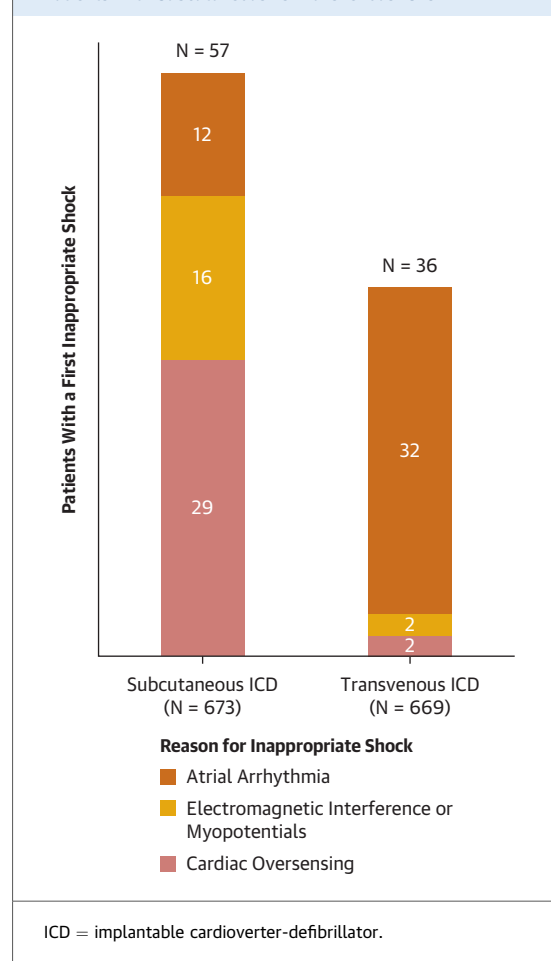


TABLE 3 Exploration of Predictors of a First Inappropriate Shock				
	Univariable Analysis		Multivariable Analysis	
	HR (95% CI)	P Value	HR (95% CI)	P Value
First inappropriate shock due to cardiac oversensing, electromagnetic interference, or myopotentials				
Randomization				
Transvenous ICD	Reference		Reference	
Subcutaneous ICD	11.76 (4.23-32.71)	<0.001	11.86 (4.27-32.98)	<0.001
Age, per 10-year increase	1.16 (0.91-1.47)	0.23		
Male sex	1.99 (0.85-4.67)	0.12	2.06 (0.88-4.83)	0.10
Secondary prevention indication	0.80 (0.39-1.65)	0.55		
Diagnosis				
Ischemic cardiomyopathy	Reference			
Nonischemic cardiomyopathy	0.64 (0.30-1.39)	0.26		
Other	0.77 (0.35-1.66)	0.50		
Left ventricular ejection fraction, per 5% decrease	1.03 (0.93-1.14)	0.56		
Beta-blocker use	0.82 (0.41-1.65)	0.58		
Antiarrhythmic drug use	0.62 (0.15-2.56)	0.51		
Trial				
ATLAS	Reference			
PRAETORIAN	0.89 (0.48-1.64)	0.70		
First inappropriate shock due to atrial arrhythmia				
Randomization				
Transvenous ICD	Reference		Reference	
Subcutaneous ICD	0.37 (0.19-0.72)	0.003	0.40 (0.20-0.80)	0.009
Age, per 10-year increase	1.05 (0.82-1.34)	0.69		
Male sex	2.15 (0.85-5.46)	0.11	2.61 (0.92-7.42)	0.07
Secondary prevention indication	1.05 (0.52-2.13)	0.89		
Diagnosis				
Ischemic cardiomyopathy	Reference			
Nonischemic cardiomyopathy	2.34 (1.25-4.40)	0.01	3.03 (1.54-5.95)	0.001
Other	0.75 (0.28-1.99)	0.57	1.38 (0.34-5.52)	0.65
Left ventricular ejection fraction, per 5% decrease	1.09 (0.97-1.23)	0.16	0.99 (0.85-1.14)	0.85
Beta-blocker use	2.13 (0.76-5.96)	0.15	3.85 (0.92-16.16)	0.07
Antiarrhythmic drug use	1.08 (0.33-3.48)	0.90		
Trial				
ATLAS	Reference			
PRAETORIAN	2.40 (1.06-5.43)	0.04	3.17 (1.04-9.70)	0.04
Abbreviations as in Table 1.				

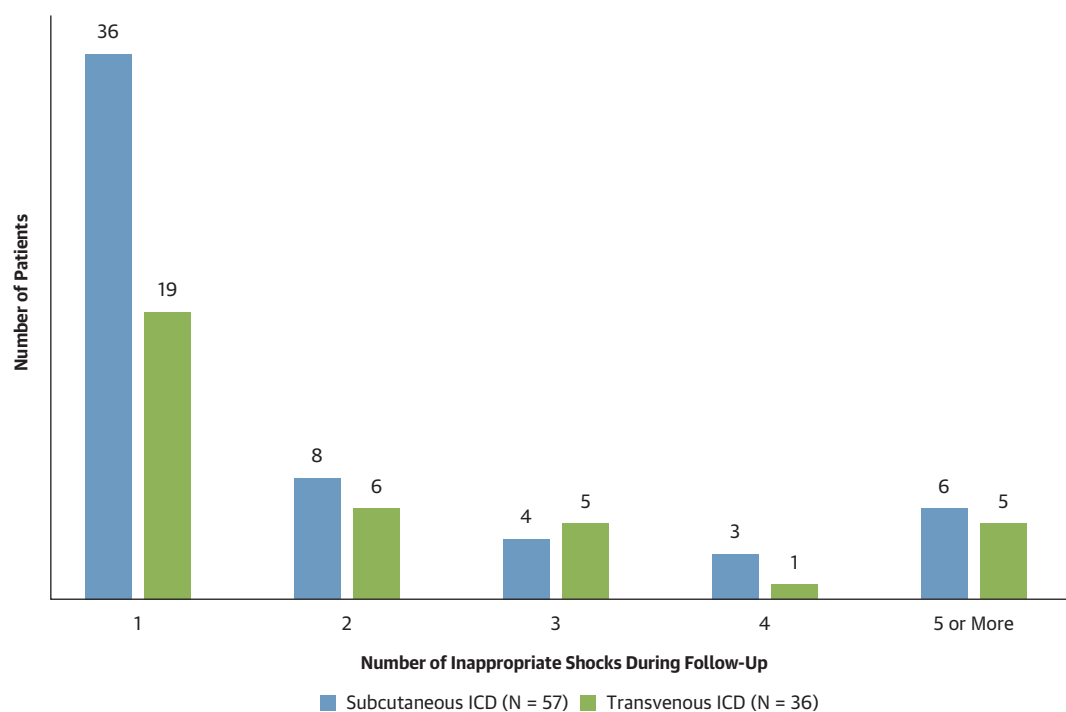
transvenous ICDs who had 5 or more inappropriate shocks during follow-up.

INAPPROPRIATE ICD THERAPY. Event rates of a first inappropriate ICD therapy (antitachycardia pacing or shock) were 2.6 vs 2.2 per 100 patient-years in patients randomized to subcutaneous ICDs and transvenous ICDs, respectively (HR: 1.16; 95% CI: 0.80-1.69; *P* = 0.44) (Supplemental Figure 3). The cumulative incidence of inappropriate ICD therapy was 6.5% vs 6.3% at 2 years and 9.9% vs 8.5% at 4 years following device implantation. The relative hazards for a first inappropriate ICD therapy due to cardiac oversensing and electromagnetic interference or myopotentials were similar to those for a

first inappropriate shock. For a first inappropriate ICD therapy due to atrial arrhythmia, however, there was an additional excess of antitachycardia pacing in patients randomized to a transvenous ICD (HR: 0.29; 95% CI: 0.16-0.53; *P* < 0.001) (Supplemental Table 5).

SUBGROUP ANALYSES. For the primary endpoint of first inappropriate shock, comparing subcutaneous vs transvenous ICDs, there was no evidence of a differential effect by age (Supplemental Figure 4), sex (Supplemental Figure 5), left ventricular ejection fraction, or baseline use of a beta-blocker and/or an antiarrhythmic drug (Table 4). A similar analysis for the secondary endpoint of first inappropriate ICD

FIGURE 4 Burden of Inappropriate Shock in Patients With at Least 1 Inappropriate Shock



The majority of patients in both groups did not have recurrent inappropriate shocks during the remainder of follow-up. The numbers of inappropriate shock in the subset of patients with 5 or more events were as follows: subcutaneous implantable cardioverter-defibrillators (ICDs): 6, 7, 10, 11, 14, and 18; transvenous ICDs: 6, 7, 9, 12, and 54.

therapy did not suggest effect modification according to the same subgroups (Supplemental Table 6).

SMART PASS ALGORITHM. Among patients randomized to receive subcutaneous ICDs, the SMART Pass algorithm was activated at baseline in 252 (37.4%). It was activated at some point during follow-up in an additional 267 (39.7%), including in patients who had initially received first-generation devices and received second- or third-generation devices during follow-up. It was not activated at any point during follow-up in the remaining 154 (22.9%). The rates of first inappropriate shock without or before and with or after SMART Pass activation were 2.6 and 2.4 per 100 patient-years, respectively (Supplemental Table 7).

Of 57 patients in the subcutaneous ICD group who had first inappropriate shocks, 30 (52.6%) had their events before or in the absence of a first activation of the SMART Pass algorithm. In analyses adjusting for its activation at baseline or during follow-up as a time-dependent covariate, SMART Pass was not associated with a reduction in receiving a first inappropriate shock for any reason (HR: 1.15; 95% CI:

0.61-2.16; $P = 0.66$) or because of cardiac oversensing (HR: 0.80; 95% CI: 0.32-1.97; $P = 0.63$).

SENSITIVITY ANALYSES. First, analyses of time to first inappropriate ICD shock and time to first inappropriate ICD therapy in the per-protocol and as-treated populations demonstrated results that were very similar to the results of the main analysis (Supplemental Tables 8 and 9). Second, the mean time until first inappropriate shock was 1,705 days in patients randomized to subcutaneous ICDs and 1,747 days in those randomized to transvenous ICDs. The difference in restricted mean survival time was 42 days (95% CI: 1-84 days; $P = 0.047$). Third, 13 patients (1.9%) randomized to subcutaneous ICDs and 13 patients (1.9%) randomized to transvenous ICDs died during follow-up. Analyses accounting for the competing risk for death yielded results that were virtually identical to those of the main analysis (Supplemental Table 10). Fourth, an analysis excluding patients randomized to subcutaneous ICD who received first-generation devices at baseline also showed a higher hazard for first inappropriate shock with a subcutaneous compared with a transvenous

TABLE 4 Time to First Inappropriate Shock With Subcutaneous vs Transvenous ICDs (Subgroup Analyses)						
	Subcutaneous ICD (n = 673)		Transvenous ICD (n = 669)			P for Interaction
	Event Rate		Event Rate		HR (95% CI)	
Age (tertiles) ^a						0.62 ^b
Lowest	14/248 (5.6)	1.8	9/209 (4.3)	1.4	1.31 (0.57-3.04)	
Middle	17/211 (8.1)	2.3	14/227 (6.2)	1.9	1.23 (0.60-2.50)	
Highest	26/214 (12.1)	3.5	13/233 (5.6)	1.4	2.33 (1.20-4.54)	
Sex						0.98
Female	7/148 (4.7)	1.3	4/145 (2.8)	0.8	1.66 (0.48-5.67)	
Male	50/525 (9.5)	2.8	32/524 (6.1)	1.7	1.62 (1.04-2.52)	
Left ventricular ejection fraction ^c						0.41
≤35%	33/380 (8.7)	2.4	23/389 (5.9)	1.5	1.52 (0.90-2.60)	
>35%	19/176 (10.8)	3.1	8/174 (4.6)	1.3	2.31 (1.01-5.29)	
Baseline use of a beta-blocker and/or an antiarrhythmic drug						0.70
No	9/111 (8.1)	2.8	5/107 (4.7)	1.3	1.90 (0.63-5.66)	
Yes	48/562 (8.5)	2.5	31/562 (5.5)	1.6	1.56 (0.99-2.45)	

Values are n/N (%) unless otherwise indicated. Event rates are reported per 100 patient-years. HRs are based on the comparison of subcutaneous vs transvenous ICDs. ^aThe median age was 45 years in the lowest tertile, 58 years in the middle tertile, and 70 years in the highest tertile. ^bP_{int} is based on age as a continuous variable. ^cCalculations are based on available data (left ventricular ejection fraction was missing for 223 patients).

ICD = implantable cardioverter-defibrillator.

ICD (HR: 1.75; 95% CI: 1.11-2.77; *P* = 0.02) (Supplemental Table 11).

DISCUSSION

This study provides the most comprehensive assessment to date of inappropriate shocks across subcutaneous and transvenous ICD systems, using patient-level data from >1,300 participants in 2 large randomized trials. The findings clarify not only the comparative frequency of inappropriate shocks by device type but also the underlying mechanisms, patient-level burden, and clinical implications. Although the subcutaneous ICD was more likely to deliver a first inappropriate shock, the absolute risk increase was modest and similar to that of other ICD-associated adverse events, such as lead dislodgement and lead fracture, which are more common with transvenous ICDs. This meta-analysis thus provides specific and quantitative information that is essential evidence for patient counseling, device selection, and future device development. Importantly, the absolute event rates were low in both groups, but the trade-offs revealed here are central to shared decision making in contemporary ICD therapy.

In this pooled analysis, the rates of first inappropriate shock with subcutaneous and transvenous ICDs were 2.5 and 1.5 per 100 patient-years, respectively. These are among the lowest rates of inappropriate shock ever documented in a clinical trial.

A systematic review and meta-analysis of studies published between 2002 and 2015 reported that the rate of first inappropriate shock in patients undergoing ICD implantation was 6.4 per 100 patient-years, with a decline in event rates in more recent compared with earlier trials.⁸ The MADIT (Multi-center Automatic Defibrillator Implantation Trial) II trial had a rate of inappropriate shocks of 7.6 per 100 patient-years,¹⁶ while in the SCD-HeFT (Sudden Cardiac Death in Heart Failure Trial), which did not allow antitachycardia pacing programming, the rate was only 4.6 per 100 patient-years.¹⁷ The PainFree SmartShock Technology trial specifically evaluated strategies to prevent shocks and demonstrated an inappropriate shock rate of 3.7% at 2 years in patients with single-chamber ICDs. Finally, the MADIT-RIT (Reduce Inappropriate Therapy) trial evaluated 2 strategies to prevent shocks in patients with multichamber ICDs and had a rate of 1.6 per 100 patient-years in its “high-rate” arm.¹⁴ Thus, over the past 2 decades, the use of atrial leads, optimization of detection algorithms, and implementation of evidence-based device programming have reduced the rate of inappropriate ICD shock to about 2 per 100 patient-years, which is similar to the rates observed in both arms of this meta-analysis.

Our analysis can support clinicians in counseling patients with an indication for implantation of an ICD without indications for pacing. The observed cumulative incidences of inappropriate shock were 6.2% with a subcutaneous ICD vs 4.4% with a transvenous

ICD at 2 years and 9.6% vs 6.3% at 4 years following device implantation. These estimates represent absolute risk measures that may inform shared decision making. Exploratory multivariable analyses did not identify consistent patient-level predictors of inappropriate shock aside from device type. At the same time, our findings should be interpreted in the context of other factors influencing the choice of ICD system, such as the anticipated risk for lead-related complications, economic considerations, and patient preferences. In numerical terms, the 1% per year increase in inappropriate shock with the subcutaneous ICD in this analysis is similar in magnitude to the rate of transvenous ICD lead dislodgement⁷ and lower than the long-term risk of transvenous ICD lead failure, both of which typically require repeat surgery.^{18,19} Furthermore, the majority of patients who had first inappropriate shocks with either device type did not have recurrent inappropriate shocks during the remainder of follow-up (63.2% vs 52.8%, respectively). Finally, it is noteworthy that the study population of this pooled analysis differs from a more general population of patients with indications for implantation of an ICD for primary prevention. Patients enrolled in the PRAETORIAN and ATLAS trials were generally younger (mean age 58.2 years) and more likely to have genetic arrhythmic syndromes, hypertrophic cardiomyopathy, or idiopathic ventricular fibrillation, and 22.3% had secondary prevention indications. These younger patients who are more likely to have preserved left ventricular systolic function typically have a higher rate of long-term lead failure.³

Another key finding of this pooled analysis is that we documented important differences in the reasons for a first inappropriate shock between subcutaneous and transvenous ICDs. The increase in first inappropriate shocks in patients with subcutaneous ICDs was driven by cardiac oversensing and electromagnetic interference or myopotentials. In contrast, the risk for a first inappropriate shock due to atrial arrhythmia was lower in patients receiving subcutaneous ICDs. This may be explained by differences in sensing (surface electrocardiography-like signals with subcutaneous vs intracardiac electrograms with transvenous ICDs) and processing of recorded signals between ICD system types.

During the conduct of the PRAETORIAN and ATLAS trials, the SMART Pass algorithm, whose aim is to reduce cardiac oversensing, became available for patients who received second- or third-generation subcutaneous ICDs.¹⁵ Previous observational studies suggested that SMART Pass was associated with a striking reduction in inappropriate shock.^{15,20,21}

However, our analysis did not detect a significant reduction. This is likely due to low statistical power, as the observed rate of first inappropriate shock was much lower than in earlier subcutaneous ICD registries. In addition, many inappropriate shocks occurred before SMART Pass was activated, limiting the ability to detect benefit. Although its high-pass filter may avoid shocks due to cardiac oversensing, SMART Pass is not expected to reduce inappropriate shocks due to electromagnetic interference or myopotentials.¹⁵ In the ATLAS trial, 63% of shocks due to electromagnetic interference were due to the use of transcutaneous electric nerve stimulation,⁷ highlighting the potential for a further reduction in inappropriate shock through patient education.

Finally, our analysis does not suggest an important difference in the risk for receiving a first inappropriate ICD therapy (antitachycardia pacing or shock) between a subcutaneous ICD (2.6 per 100 patient-years) and a transvenous ICD (2.2 per 100 patient-years). This is driven by the risk for inappropriate antitachycardia pacing in the transvenous ICD group. This finding from our pooled analysis adds to the results of a previous publication from the PRAETORIAN trial.²² Although antitachycardia pacing is typically painless and better tolerated by patients, data from MADIT-RIT and MADIT With Cardiac Resynchronization Therapy suggest a possible deleterious association between inappropriate antitachycardia pacing and mortality.^{23,24} As well, inappropriate therapies of any kind require follow-up, device reprogramming and increase clinic work load.

STUDY LIMITATIONS. First, despite pooling individual participant data from 2 randomized trials, the sample size and the number of patients with a first inappropriate shock were still modest.

Second, more than 1 in 4 patients randomized to subcutaneous ICDs received first-generation models. The SMART Pass algorithm that was developed to reduce the risk for inappropriate shock due to cardiac oversensing in the second- and third-generation models became available during the conduct of the 2 trials. Because of limited statistical power, we are unable to reliably assess the association of SMART Pass activation with the risk for a first inappropriate shock in subcutaneous ICD recipients.

Third, the programmed cutoff for the fast ventricular tachycardia zone in study participants with primary prevention indications in PRAETORIAN was lower than that informed by the MADIT-RIT results, which were not available at the time of trial initiation.^{11,14} It is possible that a higher cutoff could have further reduced the rate of first inappropriate shock

in both patients with subcutaneous and transvenous ICDs.

CONCLUSIONS

Although first inappropriate shocks were infrequent overall, they were more common with subcutaneous ICDs. First inappropriate shocks due to cardiac oversensing and electromagnetic interference were more common with subcutaneous ICDs, while first shocks due to atrial arrhythmia were more common with transvenous ICDs.

DATA AVAILABILITY. The data supporting this study may be made available to other researchers upon reasonable written request to the PRAETORIAN and ATLAS steering committees.

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APPENDIX For supplemental figures and tables, please see the online version of this paper.